



CSE6324-SOFTWARE VERIFICATION & VALIDATION

Topic: Hospital Management System

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Test Incident Report

Hospital Management System

(HMS)

Independent Verification and Validation

Project Title: HMS Independent Verification and Validation	
Date: 03/06/2026	Test Incident Report ID: HMS_TIR_1_1.0.0

Document Control

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Name	Position	Signature
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Version History

Version	Release Date	Section	Amendments
1.0	22September,2019	All	Original Document

Distribution List

Version	Release Date	Controlled Copy No	Recipient Name	Department	Issue Date	Return Date
1.0.0	22/09/2019	01		HMS	22/09/2019	

Test Incident Report

Incident report identifier	
Scope	This Test Incident Report is for incidents observed during the execution and review of follow up functional testing of HMS related to User Authentication, Patient Functions, Doctor Functions and Admin Functions. Specific information about failed or unexpected test cases is recorded in detailed incident records.
References	HMS_SRS; HMS_TP; HMS_TDS; HMS_TC; HMS_TPS; HMS_TL; Katalon Studio execution evidence; Kishan0725 Hospital Management System GitHub repository.

Test Incident Number:	TIR-02-001
Function:	F002 Login
Summary:	Duplicate Gmail registration causes patients login failed. Register with same Gmail address but unable to log in with same Gmail address.
Date and Time Incident:	11/06/2026, 3:45 PM
Context:	Patient Login / User Authentication Module
Description of Incident	
Test Procedure:	TPS-02-001 - F002 Login Test Procedure
Test Data:	Email: kai@gmail.com; Password: encrypted value recorded in Katalon; duplicated Gmail registration condition.
Expected Result:	The system should not allow to use existing Gmail to register during account creation.if gmail is valid and unique,patient should be able to login and redirect to patient dashboard.if login fail,system should show error message.
Actual Result:	The system allows multiple registrations using the same gmail.After login, it does not redirect to the patient dashboard and Katalon fails the verification

	command as the text "Welcome" isn't seen.
Unexpected Outcome:	The authentication workflow becomes inconsistent after duplicate Gmail registration. The user cannot confirm successful login although registration records exist.
Procedure to reproduce the incident:	1. Launch Google Chrome and open HMS on localhost. 2. Register a patient account using an existing Gmail address. 3. Open the patient login page. 4. Enter the duplicated Gmail address and password. 5. Click the Login button. 6. Observe that the expected dashboard welcome text is not displayed and Katalon fail at <code>verifyTextPresent("Welcome", false)</code>.
Test Environment:	Windows 11, Google Chrome, Local XAMPP Server, Apache, PHP, MariaDB or MySQL, Katalon Studio.
Attempt to repeat:	The test was repeated through Katalon Studio and the same login verification failure was observed.
Tester's Name:	LIM KAI SHEN
Observer's Name (Witness):	PANG TIAN YOU
Status of Incident	
☒ Open ☐ Assigned for Resolution ☐ Retested with the fix confirmed ☐ Approved for Resolution ☐ Fixed	
Impact	
☐ Mission Critical: Application will not function or system fails ☒ Major: Severe problems but possible to work around ☐ Minor: Does not impact the functionality or usability of the process ☐ No impact: No incident observed	
Priority	
☒ Immediate: Must be fixed as soon as possible ☐ Delayed: System is unstable but incident must be fixed before next level of test ☐ Deferred: Defect can be left in if necessary due to time or costs ☐ Not applicable: No corrective action required	
Description of the Corrective Action	
The registration module should enforce unique Gmail/email validation before inserting a new patient record. A database-level unique constraint should also be added for the patient email field. Any duplicate email records need to be deleted	

and the login authentication query needs to be reviewed to ensure a single patient account identifies one email.

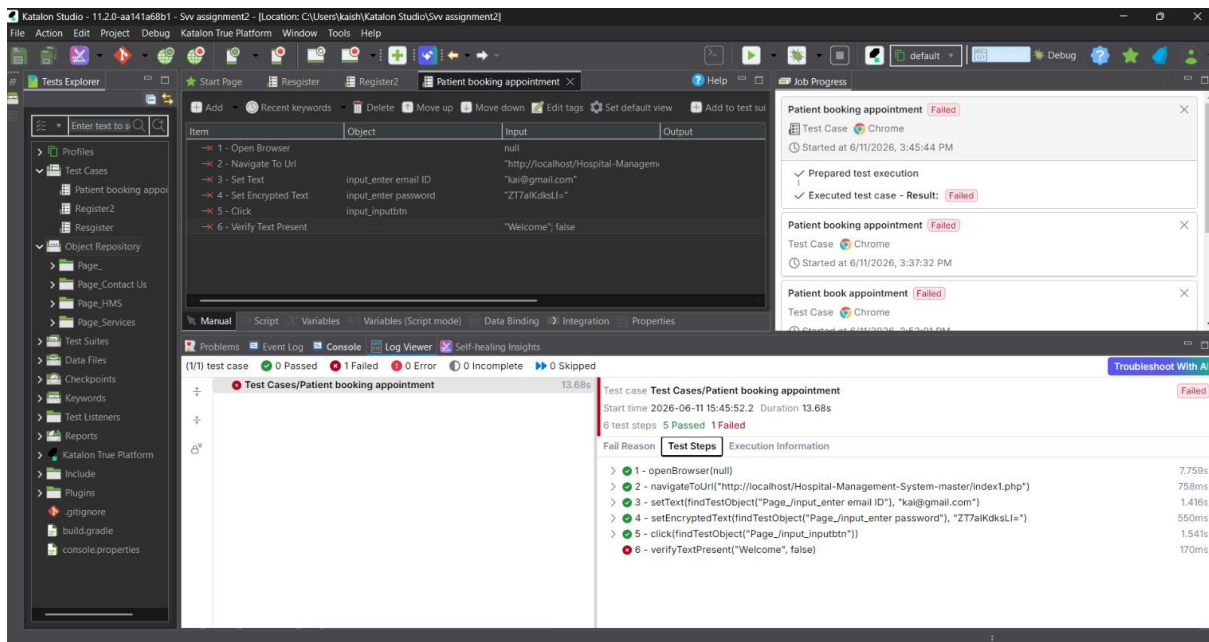


Figure 1: Katalon execution evidence for TIR-02-001

Test Incident Number:	TIR-04-001
Function:	F004 Book Appointment
Summary:	After register direct book appointment will fail because the system trigger database insertion error and does not display the expected booking success confirmation.
Date and Time Incident:	11/06/2026, 4:00 PM
Context:	Patient Module / Book Appointment Page
Description of Incident	
Test Procedure:	TPS-04-001 - F004 Book Appointment Test Procedure
Test Data:	Specialization: General; Doctor: Dinesh; Date: 2026-06-17; Time: 12:00:00; Patient account created in the same workflow.
Expected Result:	Appointment Details entered by user should get validated by system and the appointment gets created along with showing the "Appointment successfully booked" success message.
Actual Result:	After submit the valid appointment detail, did not show the success

	message. Katalon fail at verifyTextPresent("Appointment successfully booked", false). Additionally during this failure area a fatal mysqli_sql_exception occurs while attempting to insert an appointment in the Test Log.
Unexpected Outcome:	The patient cannot complete the core booking workflow immediately after registration. The appointment record is not confirmed to the user and the booking process becomes unreliable.
Procedure to reproduce the incident:	1. Launch Google Chrome and open HMS on localhost. 2. Register or log in as a patient. 3. Go to the Book Appointment page. 4. Select Specialization = General. 5. Select Doctor = Dinesh. 6. Enter Date = 2026-06-17. 7. Select Time = 12:00:00. 8. Click the create new entry button. 9. The appointment was booked successfully message "Appointment successfully booked" does not show up and verification step in Katalon fails.
Test Environment:	Windows 11, Google Chrome, Local XAMPP Server, Apache, PHP, MySQL or MariaDB, Katalon Studio.
Attempt to repeat:	Use Katalon Studio to repeat the test. The booking verification failed again when the expected success text was not found.
Tester's Name:	LIM KAI SHEN
Observer's Name (Witness):	PANG TIAN YOU
Status of Incident	
☒ Open ☐ Assigned for Resolution ☐ Retested with the fix confirmed ☐ Approved for Resolution ☐ Fixed	
Impact	
☒ Mission Critical: Application will not function or system fails ☐ Major: Severe problems but possible to work around ☐ Minor: Does not impact the functionality or usability of the process ☐ No impact: No incident observed	
Priority	
☒ Immediate: Must be fixed as soon as possible ☐ Delayed: System is unstable but incident must be fixed before next level of test	

☐ Deferred: Defect can be left in if necessary due to time or costs

☐ Not applicable: No corrective action required

Description of the Corrective Action

The appointment booking backend should be reviewed, especially the database insertion logic and the patient session or patient ID used after direct registration. The system should handle database exceptions safely, validate that the patient ID exists before booking, create the appointment record correctly, and display the success message only after a successful insert.

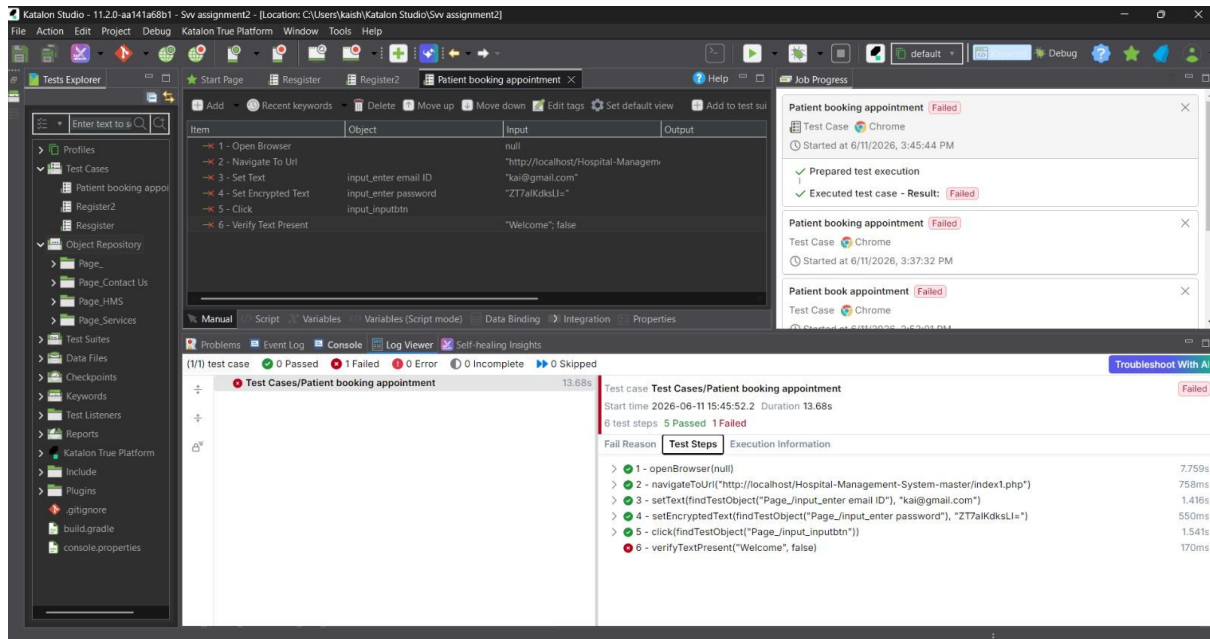


Figure 2: Katalon execution evidence for TIR-04-001

Test Incident Number:	TIR-10-001
Function:	F010 Prescribe Medication
Summary:	Doctor can repeatedly prescribe medication for the same appointment. Multiple records for a single Appointment ID is not prevented.
Date and Time Incident:	11/06/2026, 4:40 PM
Context:	Doctor Module / Prescribe Medication Interface
Description of Incident	
Test Procedure:	TPS-10-001 - F010 Prescribe Medication Test Procedure
Test Data:	Disease: good; Allergy: good; Prescription: good; same appointment selected more than once.
Expected Result:	If there is an existing prescription for the appointment, then it should be unable to create a new prescription or show a

	error message, "Prescription already issued for this appointment".
Actual Result:	The doctor could successfully reopen the prescribed action and write another prescription for that particular appointment. Katalon failed at <code>verifyTextPresent("Prescription already issued for this appointment", false)</code> because the expected duplicate-prevention message was not displayed.
Unexpected Outcome:	The system allows duplicate medical prescription records for the same appointment, causing inconsistent clinical records and possible repeated bill or prescription information.
Procedure to reproduce the incident:	1. Launch Google Chrome and open HMS on localhost. 2. Log in using a valid doctor account. 3. Navigate to the appointment list. 4. Select an active appointment and click Prescribe. 5. Enter disease, allergy, and prescription details. 6. Submit the prescription. 7. Return to the appointment list. 8. Click Prescribe again for the same appointment. 9. Verify the system allow perform a second prescribe action without display "Prescription already issued for this appointment".
Test Environment:	Windows 11, Google Chrome, Local XAMPP Server, Apache, PHP, MySQL / MariaDB, Katalon Studio.
Attempt to repeat:	Use Katalon Studio repeat the test. The same duplicate prescription behaviour was observed, and the expected prevention message was not found.
Tester's Name:	LIM KAI SHEN
Observer's Name (Witness):	PANG TIAN YOU
Status of Incident	
☒ Open ☐ Assigned for Resolution ☐ Retested with the fix confirmed ☐ Approved for Resolution ☐ Fixed	
Impact	
☐ Mission Critical: Application will not function or system fails ☒ Major: Severe problems but possible to work around	

- ☐ Minor: Does not impact the functionality or usability of the process
- ☐ No impact: No incident observed

Priority

- ☒ Immediate: Must be fixed as soon as possible
- ☐ Delayed: System is unstable but incident must be fixed before next level of test
- ☐ Deferred: Defect can be left in if necessary due to time or costs
- ☐ Not applicable: No corrective action required

Description of the Corrective Action

The prescription module should check to see whether a prescription record already exists for the selected Appointment ID prior to adding the new prescription information. Once prescription is made, the “Prescribe” button should be disabled, hidden or read-only view. To prevent a duplicate prescription record on the same appointment: Add a backend validation rule and database unique constraint.

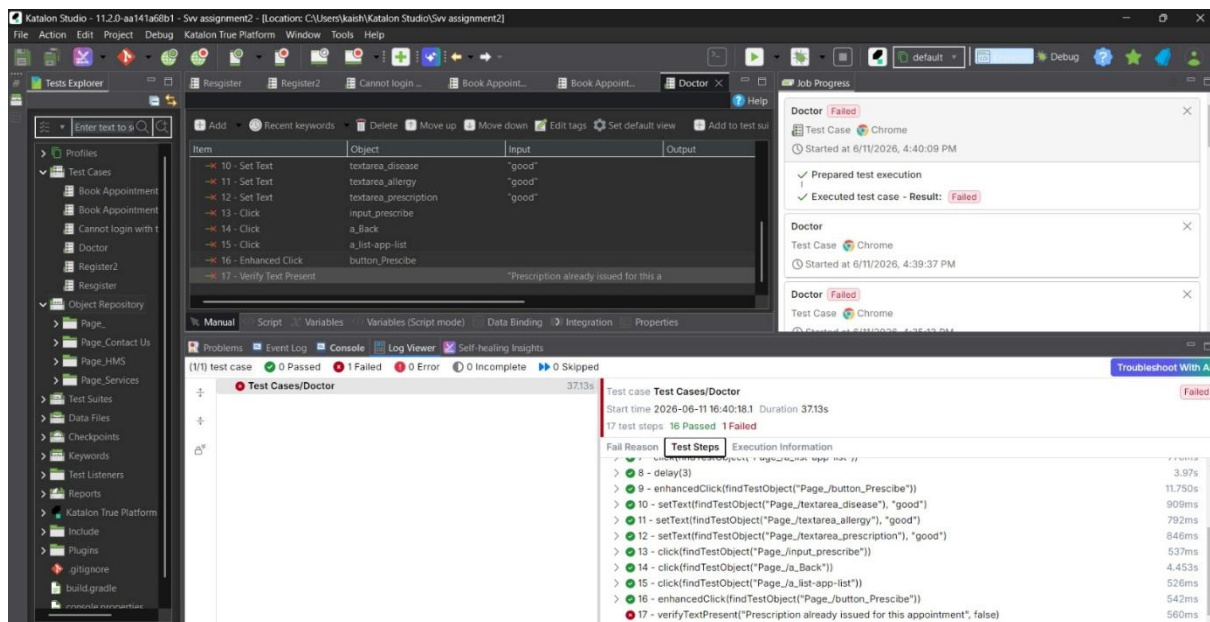


Figure 3: Katalon execution evidence for TIR-10-001

Conclusion and Recommendation

Based on the test execution, three incidents were recorded for the Hospital Management System. The incidents involve the authentication workflow, appointment booking workflow, and prescription workflow. The most serious issue is the appointment booking incident because the patient booking process failed instead of producing the expected success result. The duplicate prescription incident is also significant because it affects the integrity of medical records. The login incident should be corrected because duplicate Gmail handling can block patient access and create inconsistent user records.

The development team is highly recommended to fix the above mentioned issues before next test round. After all necessary changes are implemented, it is essential

to perform regression testing for F001 Register Account, F002 Login, F004 Book Appointment and F010 Prescribe Medication.

Approvals

Name	Job Title	Signature
CHIN ZHEN HO, PANG TIAN YOU, LIM KAI SHEN	Test Analyst	CHIN,PANG,LIM
CHIN ZHEN HO	Test Lead	CHIN
Ms. Venushini A/P Rajendran	Test Manager	